

1. With regard to the NIST 800-131/2010 document, which are the recommended minimal levels of security for ES? ([SP 800-131 \(June 11 2010\): Recommendation for the Transitioning of Cryptographic Algorithms and Key Lengths](#))

- a. Until 2010 – DSA : L=1024 bits; N=160 bits; RSA: K=1024 bits; ECDSA: f=160
- b. Until 2030 – DSA : L=4096 bits; N=256 bits; RSA: K=4096 bits; ECDSA: f=256
- c. Until 2010 – DSA : L=2048 bits; N=196 bits; RSA: K=2048 bits; ECDSA: f=196
- d. Until 2030 – DSA : L=2048 bits; N=224 bits; RSA: K=2048 bits; ECDSA: f=224

- **DSA (Digital Signature Algorithm):**

- L = the length of the prime modulus
- N = the length of the cryptographic hash

- **RSA:** Uses a key size (K) which represents the modulus length.

- **ECDSA (Elliptic Curve Digital Signature Algorithm):** f = the size of the curve's parameters

- **Timelines:** The key concept in NIST 800-131A is the transition. There are dates after which certain key lengths are disallowed or deprecated.

2. About public key Cryptographic Systems, we can say that they:

- a. Are symmetric
- b. Use different keys, that are math linked, for encrypting and decrypting
- c. Use different keys, randomly chosen, for encrypting and decrypting
- d. Are based on trap-door, one-way function

3. The RSA system uses:

- a. Operations with large integer number
- b. One key, publicly available
- c. One public key, one private key
- d. Operation on elliptic curves

4. The revocation of certificate is done:

- a. When the public key inside the certificate is compromised
- b. When the validity of the certificate has expire
- c. By publishing the certificate according to the OCSP protocol
- d. When the private key is compromised

5. Which is the role of HSM (Hardware Security Mode)?

- a. Keeps the used cryptographic algorithm secret
- b. Assures the protection of the cryptographic keys
- c. Accelerate the cryptographic operations
- d. Assures the protection of the computer which the CA software is installed

6. The certificate Policy –CP

- a. Describes the security which governs a CA
- b. Describes in detail the mechanism and procedures of the CA
- c. Describes the registration process of the users
- d. Represents an operational manual for the CA

7. The law regarding the ES adopted by Parliament of Romania, in July 2001, establish :

- a. The fees for providing CA
- b. The legislative environment to electronic documents
- c. The conditions for providing electronic signatures certificate services
- d. The conditions for the ES of documents, in electronic communication

8. The extended electronic signature represents an electronic signature which complies with the following conditions:

- a. It is uniquely linked to the signer, assuring his identification
- b. Is created using specific hardware devices (smart cards or tokens)
- c. It is linked to the electronic document being signed
- d. It is created with means owned exclusively by signer

9. RSA is a cryptographic algorithm based on:

- a. Difficulty of computing discrete logarithms in finite field
- b. Fermat's theorem
- c. Difficulty of computing exponents in finite field
- d. Difficulty of large integer numbers factorization

10. About time-stamping, we can say that:

- A. the RFC 3161 – Internet X.509 PKI Time Stamp Protocol (TSP), represents its standard environment
- B. is accomplished by a certificate service provider
- C. it's generated by the application and counter-signed by a time-stamping service provider
- D. the law on time-stamping, adopted in Romania, in 2004, represents its legislative environment

11. Electronic document is equivalent, in condition of effect, with written documents if:

- a. It has an extended ES associated
- b. The signature is based on a qualified article
- c. The signature is created using a secure mechanism for creating digest signature
- d. The certificate service provider is accredited

12. The effort for breaking a symmetric system with a 128 bits key, with a specialized system, using brute force attack, is around:

- a. 10^{18} years
- b. 3.6 hours
- c. 18^{18} years
- d. 7000 years

13. DSA is a standard which:

- A. was created by Rivest-Shamir and Adleman
- B. is used for encrypting data
- C. can be implemented on elliptic curves
- D. is used only for digital signature

14. Diffie-Hellman key distribution scheme, with 3 users. The common values are the base, $a=2$ and the modulus, $p=7$. User A has $\text{PubKXA} = 3$ and computes with Priv KYA . User B has $\text{PubKXB}=4$ and computes with Priv KYb . User C has $\text{PubKXC}=5$ and computes with Priv KYc . The 3 users exchange their Public keys. Compute the secret keys used to encrypt the traffic between users.

- a. $KAB=1$, $KAC=4$, $KBC=1$
- b. $KAB=4$, $KAC=1$, $KBC=1$
- c. $KAB=4$, $KAC=1$, $KBC=4$
- d. $KAB=1$, $KAC=1$, $KBC=4$

15. Which solution(s) was/were proposed for satisfying the WYSIWYG request? (what you see is what you get)

- A. the signing of a graphic representation of the document
- B. the use of a secure display program, for viewing the documents
- C. the use of logos for certifying the authenticity of the document
- D. the use of a document format, which has an unique graphic representation

16. If, at the moment of validating a signature, the signing certificate is revoked, the signature may be declared valid only if:

- A. the signed document was time-stamped after being stamped and before the revocation of the certificate**
- B. the signed document was time-stamped before being signed
- C. the signature will be declared invalid anyway, because the certificate is revoked when the validations is done
- D. the certificate of the signer hasn't expired at the moment of validation

17. The secured mechanism for creating electronic signatures must satisfy the following conditions:

- A. the signature creation data must be protected effectively by the signer against their use by unauthorized persons
- B. it must not modify the electronic document which will be signed and not prevent its view by the signer, before finishing the signing process
- C. the signature creation data must appear only once and its confidentiality must be assured
- D. the signature creation data must not be deduced and the signature must be protected against forgery

18. Which of the following devices are specific for Secure Signature Creation Devices (SSCD)?

- A. Smart cards
- B. USB tokens
- C. PCMCIA
- D. Biometric readers

19. The electronic signature assures:

- A. the authentication of the data link between sender and receiver
- B. the authentication of the signed message (its integrity verification)
- C. the authentication of the sender of the signed message
- D. the authentication of the receiver of the signed message

20. The OCSP server, in order to obtain revocation information, may:

- A. process CRLs, issued by the Certification Authority
- B. call the services of another OCSP server, authoritative for the PKI domain in which the certificate was issued
- C. use SSL protocol
- D. use LDAP v2 protocol

21. The law no. 589/2004, regarding the legislative environment for the electronic notary activity states that:

- A. the electronic notary closure contains as signatures: the extended electronic signature of the parties, the extended electronic signature of the public notary, the extended electronic signature of the notary bureau
- B. the electronic notary closure contains as signatures: the extended electronic signature of the parties, the extended electronic signature of the notary bureau
- C. the public notary is obliged to keep the electronic archive of electronic notary documents for 20 years
- D. the requests for an electronic notary document shall be handed to the public notary in electronic form, signed with the extended electronic signature of the requester

22. Let there be a RSA protocol for transporting the key, with 2 users. User A publishes his public key $[e_A=3, n_A=55]$ and computes his private key $[d_A=27, n_A=55]$. Let us suppose that user B wants to issue an encrypted message towards user A. User B envelopes the session key ($K_{AB}=7$), using the public key of user A: $K' = (K_{AB})^{e_A} \bmod n_A = 7^3 \bmod 55 = 13$. Please show the way in which user A determines the session key

($K_{AB}=7$), using his private key and the received enveloped key ($K'=13$).

- A. $K_{AB} = (K')^{d_A} \bmod n_A = (13)^{27} \bmod 55$
- B. $K_{AB} = (K')^{n_A} \bmod e_A = (13)^{55} \bmod 3$
- C. $K_{AB} = (K')^{e_A} \bmod n_A = (13)^3 \bmod 55$
- D. $K_{AB} = (K')^{n_A} \bmod d_A = (13)^{55} \bmod 27$

23. The Windows operating system, in order to build the certificate chain, it automatically discovers the necessary certificates in the following certificate stores:

- A. „Personal”
- B. „Intermediate Certification Authorities”
- C. „Enterprise Trust”
- D. “Trusted Root Certification Authorities”

24. According to the time stamping law, the time stamp information contains, among other data:

- a. the extended electronic signature of the author of the electronic data
- b. the electronic data or a hash function of it
- c. the extended electronic signature of the time-stamping issuing authority
- d. date and time

25. A variation of the Diffie Hellman scheme, for signature is:

The public elements: q (prime number), $a < q$ (prime root of $q-1$)

Private key: X , $X < q$

Public key: $Y = a^X \bmod q$

The message to be signed M , and its hash $h(M)$

The signature: a^Z , where Z satisfies the equation $Z * h = X \pmod{q-1}$

Show the way in which the verification of the signature is done:

- a. $Y = (a^Z)^M = a^X \bmod q$
- b. $Y = (a^M)^h = a^X \bmod q$
- c. $Y = (a^Z)^h = a^X \bmod q$
- d. $Y = (a^Y)^h = a^X \bmod q$

26. In Chaum blind signature scheme, for (blinding) A computers: $m^* = mk \bmod n$. indicate the signing and un-blinding mechanism:

- A. (un-blinding) A computes: $s^* = (m^*)^d \bmod n$
- B. (signing) B computes: $s = k^{(-1)} s^* \bmod n$
- C. (un-blinding) A computes: $s = k^{(-1)} s^* \bmod n$
- D. (signing) B computes: $s^* = (m^*)^d \bmod n$

27. The correct equation, which has to be verified for validating the signature, is

In Shamir's identity based signature scheme, in which (e, d) are the RSA master keys, the signing (s, t) has the following steps:

- it randomly chooses r ;
- it computes $t = r^e \bmod n$;
- it computes $f = H(t, m)$;
- it computes $sID = ID^d \bmod n$;
- it computes $s = sID * r^f \bmod n$.

- A. $s^e = ? = ID * t^H(t, m) \bmod n$
- B. $s^e = ? = H(t, m) * t^ID \bmod n$
- C. $s^e = ? = H(m) * t^ID \bmod n$
- D. $s^e = ? = t^H(t, m) \bmod n$

28. About blind signatures, we can say that:

- a. they were theoretically developed by Chaum
- b. they were theoretically developed by Chaum-van Antwerpen
- c. they were initially designed as a protocol for securings web apps
- d. they are used in the field of electronic cash

29. MQV (Menezes -Qu-Vanstone) is

- a. an authenticated protocol for establishing a key, based on Diffie-Hellman
- b. an authentication protocol based on Diffie-Hellman
- c. based on a pair of keys: a static one and an ephemeral one
- d. a distribution key protocol, based on Diffie-Hellman

30. DSA is a cryptographic algorithm based on:

- a. the difficulty of multiplying large prime numbers
- b. the difficulty of computing exponents in finite field
- c. the difficulty of large integer numbers factorization
- d. the difficulty of computing discrete logarithms in finite field

31. Electronic signatures are based on:

- a. scanned handwritten signatures
- b. cryptographic systems with public keys and hash functions
- c. biometric data (finger print, iris, etc.)
- d. cryptographic systems with secret keys and hash functions

32. Signing a document is done using:

- a. the private key of the private message sender
- b. the private key of the message recipient
- c. the sender's public key message
- d. the public key of the recipient

33. Signature Verification of a document is done using:

- a. the private key of message sender
- b. the private key of the message recipient
- c. the public key of the sender
- d. the public key of the recipient

34. For electronic signature can be used the next cryptographic algorithms:

- a. RSA, DSA, DSA over elliptic curves
- b. SHA-1
- c. AES over elliptic curve
- d. DES

35. XML signatures are:

- a. data structures containing the signature value and signature algorithms used
- b. data structures containing the signature value and the necessary data for the verification process
- c. at the enveloped XML signatures, the signed object encapsulates signature
- d. at the enveloped XML signatures, the signed object encapsulates signature

36. Time stamping

- a. it is realized by the the application and countersigned by a time stamping service provider
- b. it is legal regulated by time stamping law enacted in Romania in 2004
- c. it is legal regulated by RFC-3161 Internet X.509 PKI Time Stamp Protocol (TSP);
- d. is performed by a certification service provider

37. RFC 2587 (Internet X.509 PKI Operational Protocols - LDAPv2):

- a. defines the access method to the certificates repository
- b. is a PKCS standard developed by RSA Data Security for access to storage repository
- c. defines the attributes and object classes supported by the LDAP server
- d. defines the method of online checking certificates

38. For an RSA cryptographic system with public key ($e = 5$, $n = 21$), the pair private key (d , n) of this key is:

R: $d = \underline{\hspace{2cm}}$, $n = \underline{\hspace{2cm}}$

39. Among static biometric methods are:

- a. fingerprint
- b. Hand geometry
- c. identification of the retina
- d. Voice recognition

40. Main current issues concerning the use of biometric signature methods are:

- a. the reliability of biometric identification methods
- b. lack of appropriate standard
- c. lack of smartcards with biometric sensors
- d. the impossibility of memorizing biometric characteristics due to lack of space on card

41. The electronic signatures Directive issued by the Commission of the European Communities the subject of debate and adoption in the Parliament and the Europe Council in 1999 refers to:

- a. the definition of terminology used in electronic signature
- b. the legal effects of electronic signatures
- c. international aspects of using electronic signatures
- d. the requirements for certification service providers and how their certification and authorization

42. A secure mechanism to create electronic signature must meet the following conditions:

- a. not to change the content of the document that must be electronically signed and shall not prevent it to be presented to signer prior to the completion of signature
- b. signature creation dates can not be deducted and the signature to be protected against forgery
- c. signature creation dates can not occur only once and their confidentiality to be assured
- d. signature creation dates to be effectively protected by the signatory against their use by unauthorized persons

43. Accreditation of certification service providers:

- a. is mandatory
- b. is done by the Regulatory Authority and Supervision
- c. is voluntary
- d. regularly made by surveillance procedure

44. DSA is a cryptographic algorithm based on:

- a. the difficulty of computing discrete logarithms in finite fields
- b. the difficulty of large integer numbers factorization
- c. the difficulty of computing exponents in finite field
- d. the difficulty of multiplying large prime numbers